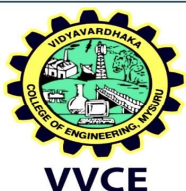


SEMESTER – VI

Course Name	: Blockchain Technology	Course Code:	BIS/CSBT6 03
Number of Lecture Hours / Week	: 03	CIE Marks:	50
Number of Tutorial / Practical Hours / Week	: 00	SEE Marks:	50
Total Number of Lecture + Tutorial/Practical Hours	: 40	SEE Duration:	03 Hrs.
L:T:P	: 3:0:0	Credits:	03
Course Prerequisites Basic knowledge of computer networks, Cryptography, and programming skills			
Course Overview The course provides a platform for students to learn about the history, important blockchain concepts, philosophy of decentralization behind blockchain and discussions happening within the blockchain environment. In addition, applications of blockchain and the impact it could have on the business world.			
Course Learning Objectives (CLOs) - To understand basic concepts of blockchain. - To provide knowledge about security mechanisms used in blockchain. - To provide hands-on experience with the concepts.			
Modules			Teaching Hours
Module 1 Blockchain: Distributed systems, CAP theorem, Byzantine Generals problems, Consensus, Introduction to blockchain, various technical definitions of blockchains, Generic elements of a blockchain, Features of a blockchain, Applications of blockchain technology, Tiers of blockchain technology, Types of Blockchain Consensus in blockchain, CAP theorem and blockchain, Benefits and limitations of blockchain. SLT: The history of blockchain Textbook: Ch1, Pg. No.: 10-32			08
Module 2 Decentralization: Decentralization using blockchain, Blockchain and full ecosystem decentralization - Smart contract, organizations, autonomous organizations, autonomous corporations, autonomous societies, Platforms. Cryptography and Technical Foundations: Mathematics, Cryptography, Cryptographic primitives: Symmetric cryptography, DES, AES SLT: Methods and Applications of decentralization Textbook: Ch2: Pg. No.: 34-49 & Ch3: Pg. No.: 50-63			08
Module 3 Cryptographic primitives: Asymmetric cryptography, Hash functions, Secure Hash Algorithms (SHAs), Merkle trees, Patricia trees, Distributed hash tables (DHTs), Digital signatures. SLT: Public and private keys- RSA Textbook: Ch3: Pg. No.: 66-70, 87-98			08



Module 4 Bitcoin: Bitcoin definition, Transactions -life cycle, structure, Blockchain: The structure of block The structure of a block header, Wallets. Smart Contracts: History, Definition, Ricardian contracts SLT: Types of transaction Textbook: Ch4: Pg. No.: 111- 129, 145-148 & Ch6	08
Module 5 Ethereum 101: Introduction - Ethereum clients and releases, Ethereum blockchain, Elements of the Ethereum blockchain, Ethereum virtual machine (EVM)- Execution Environment, Accounts, Block, Ether, Messages, Mining. SLT: The Ethereum stack Textbook: Ch7: 210-227, 235-239, 244-254	08
Textbook 1. Imran Bashir, "Mastering Blockchain", Packt, 2017	
Reference Books 1. Mastering Bitcoin: Programing the Open Blockchain Paperback-2017 by Andreas M. O'rielly 2. Arvind Narayanan, Joseph Bonneau, Edward Felten, Andrew Miller and Steven Goldfeder. Bitcoin and cryptocurrency technologies: a comprehensive introduction. Princeton University Press, 2016.	

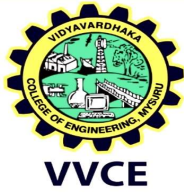
Course Outcomes (COs) At the end of the course students will be able to	
CO1	Understand the blockchain terminologies with its applications.
CO2	Examine and apply the decentralization, asymmetric cryptographic primitives, and Bitcoin concepts
CO3	Analyze the principles of Ethereum and its transactions in blockchain
CO4	Design and demonstrate a blockchain using Ethereum Tool and make effective presentation in a team.

CO - PO - PSO Matrix

CO	PO											PSO			
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2	PSO3	PSO4
CO1	2													2	
CO2	2													2	
CO3		2												2	
CO4					2									2	
Avg.	2	2			2									2	

Choice of Answering in CIE & SEE:

Module 1 & 5	Students are learning detailed basic concepts and Ethereum tools and their implementations in blockchain. Hence choice is considered for these two modules. The rest of modules contains limited concepts of decentralization, cryptography, and smart contracts for which internal choice is not preferred.
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Vidyavardhaka Sangha®, Mysore
VIDYAVARDHAKA COLLEGE OF ENGINEERING

Autonomous Institute, Affiliated to Visvesvaraya Technological University, Belagavi
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